

eira 2006]、异常检测 [Lakhina 2004; Lakhina 2005; Barford 2009] 及更多的主题正在研究中。我们将不涉及更宽泛的服务管理主题 [Saydam 1996; RFC 3052]，如带宽、服务器容量和其他为满足某企业特定任务的服务要求所需要的计算/通信资源。

一个经常被问到的问题是：什么是网络管理？我们上面的讨论是从网络管理的需求和展示几个网络管理应用的角度进行的。在这节中，我们将用一个句子（虽然它相当冗长）来概括网络管理的定义 [Saydam 1996]。

“网络管理包括了硬件、软件和人类元素的设置、综合和协调，以监视、测试、轮询、配置、分析、评价和控制网络及网元资源，用合理的成本满足实时性、运营性能和服务质量的要求。”

这句话虽绕口，但它是一个好而易于使用的定义。在后面的内容中，我们将为这个相当本质的网络管理定义增加一些内容。

9.2 网络管理的基础设施

我们在前面一节中看到，网络管理要求具有“监视、测试、轮询、配置……和控制”网络中的硬件和软件组件的能力。因为这些网络设备是分布式的，这将在最低程度上要求网络管理员能够从远程的一个实体收集数据（例如，为了监视目的），并实施远程实体的改变（例如，控制它）。一个人类的类比例子将帮助我们理解网络管理所需要的基础设施。

假定你是一个在全世界具有分支机构的大机构的领导。你的工作是确保该机构的各部分正常运转。你将如何做到这一点呢？至少，你将周期性地从这些分支机构以报告的形式收集数据，并且定量地测算各种活动、生产率和预算。你偶尔会（不会是经常）被明确地告知在这些分支机构中的某个机构存在一个问题；该分支机构经理是一个希望向公司高层爬的人（也许希望得到你的位子），他也许主动向你提供报告，指出在其领导的分支机构中工作进行得十分顺利。你会详细地审查收到的报告，希望看到各地机构正常运行，但无疑发现了需引起你重视的问题。你可能与存在问题的分支机构发起一对一对话，为了搞清楚该问题而收集更多的数据，然后向该分支机构的经理传达一个行政命令（“纠正这件事！”）。

隐含在这个非常普通的人类活动背后的是一个控制机构的基础设施，老板（你）、被控制的远程站点（分支机构）、你的远程代理（分支机构的经理）、通信协议（用于传输标准的报告和数据或一对一对话）和数据（报告内容以及各种活动、生产率和预算的定量估算）。在人类机构管理中的这些部件，每个都在网络管理中有对应的东西。

网络管理系统的体系结构在概念上等同于这个简单的人类组织的类比例子。网络管理领域对于网络管理体系结构的各种组件都有自己的特定术语，因此我们这里采用这些术语。如图 9-2 中所示，网络管理体系结构有三个重要组件：一个管理实体（在上述类比中的老板，即你），多个被管设备（分支机构）和一个网络管理协议。

管理实体（managing entity）是一个应用程序，通常有人位于其中，该程序运行在网络运营中心（NOC）的集中式网络管理工作站上。管理实体是执行网络管理活动的所在地；它控制网络管理信息的收集、处理、分析和/或显示。正是在这里，发起控制网络行为的动作并且人类网络管理员与网络设备打交道。

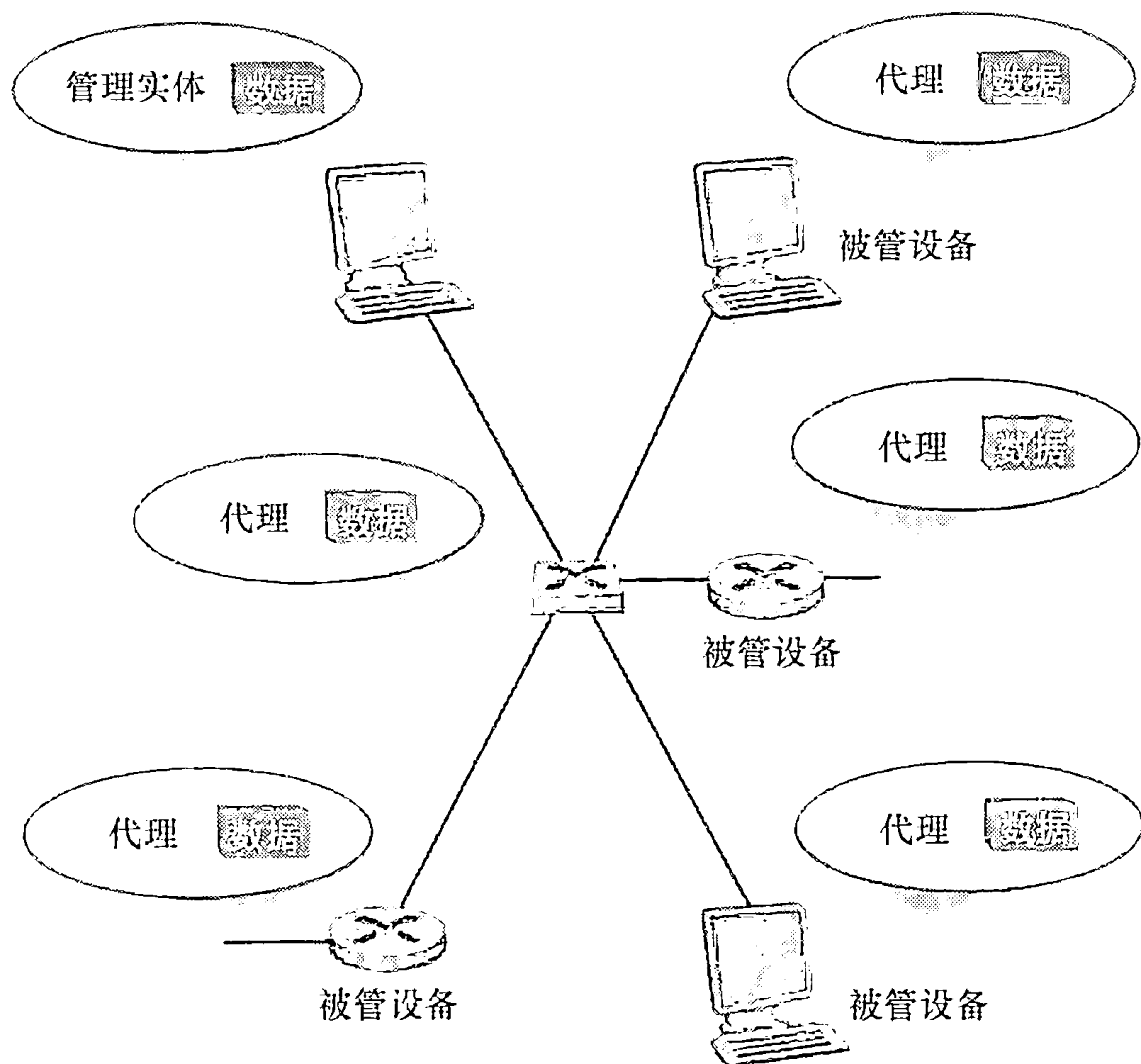


图 9-2 网络管理体系结构的主要组件

被管设备（managed device）是网络装备的一部分（包括它的软件），位于被管理的网络中。它相当于我们上述类比中的分支机构。一个被管设备可能是主机、路由器、网桥、集线器、打印机或调制解调器。在一个被管设备中，有几个所谓被管对象（managed object）。这些被管对象是被管设备中硬件的某些有效部分（如一块网络接口卡）和某些硬件与软件的配置参数集合（如 RIP 这样的域内路由选择协议）。在我们的人类的类比中，被管对象可能是分支中的一个部门。这些被管对象具有与它们相关的部分信息，这些信息收集在管理信息库（MIB）中；我们将看到，这部分信息的值可供管理实体利用（并且在许多情况下能够被它设置）。在前面的人类的类比中，MIB 对应于在分支机构和中心办事处之间交换的定量数据（活动性、生产率和预算的度量，并且管理实体能够设置 MIB）。我们将在 9.3 节中详细学习 MIB。最后，在每个被管设备中还驻留着网络管理代理（network management agent），它是运行在被管设备中与管理实体通信的进程，在管理实体的命令和控制之下，在被管设备上采取本地动作。该网络管理代理对应于上述类比中的分支机构经理。

网络管理体系结构的第三部分是网络管理协议（network management protocol）。该协议运行在管理实体和被管设备之间，允许管理实体查询被管设备的状态，并经过其代理间接地在这些设备上采取行动。代理能够使用网络管理协议向管理实体通知异常事件（如组件故障或超过了性能阈值）。重要的是注意到网络管理协议自己不能管理网络。恰恰相反，它为网络管理员提供了一种用来管理（“监视、测试、轮询、配置、分析和控制”）网络的工具。这是一种细微但却重要的区别。

尽管网络管理基础设施概念上简单，但一个人经常被网络管理行话如“管理实体”、“被管设备”、“管理代理”和“管理信息库”等搞得一头雾水。例如，用网络管理行话来说，在简单的主机监视场景下，位于“被管设备”中的“管理代理”由“管理实体”周

期性地查询，这样一个简单的想法说起来却十分绕口。幸运的是，当我们继续研究本章内容的时候，心中想着上述人类组织机构的类比例子——这个类比例子与网络管理有的明显对应，将有助于我们的学习。

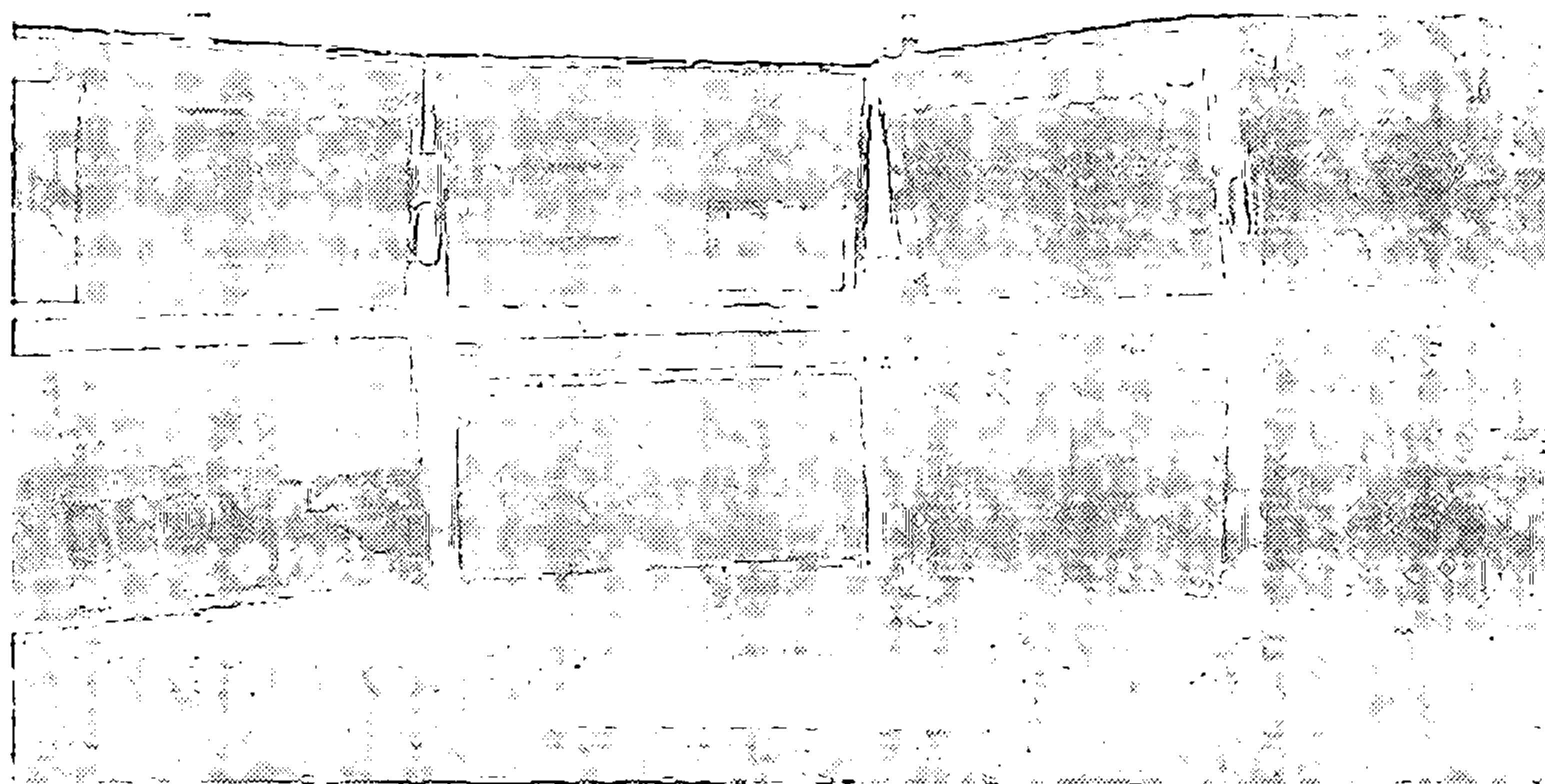
上述有关网络管理体系结构的讨论具有一般性，能够宽泛地应用于一些网络管理标准和过去若干年提出的方法中。网络管理标准成熟于 20 世纪 80 年代后期，这时推出了 OSI CMISE/CMIP（Common Management Information Services Element/Common Management Information Protocol，公共管理信息服务元素/公共管理信息协议）[Piscatello 1993; Stallings 1993; Glitho 1998] 和因特网 SNMP（Simple Network Management Protocol，简单网络管理协议）[RFC 3410; Stallings 1999; Rose 1996] 两个重要的标准 [Subramanian 2000]。这两个标准的设计独立于特定厂商的产品或网络。因为 SNMP 迅速设计和实施恰逢网络管理需求变得十分强烈之时，所以它得到广泛的使用和接受。今天，SNMP 已经成为应用和实施最为广泛的网络管理框架。我们将在下面内容中详细学习 SNMP。

实践原则

Comcast 的网络运营中心

Comcast 基于光纤的世界级 IP 网络向 4900 万个综合视频、数据和语音客户传送汇合的产品和服务。Comcast 网络包括超过 618 000 英里的建设线路、138 000 英里的光纤线路、30 000 英里的主干、122 000 个光结点以及用于 Comcast 内容交付网络的巨量存储，该 CDN 传递了超过 134TB 的按需视频产品。Comcast 网络的每个部分，甚至包括顾客的家庭或工商企业，都由该公司的运营中心之一进行监视。

Comcast 运营两个国内网络运营中心，这些中心管理国内主干、区域网络、国内应用以及支持语音、数据和视频基础设施的特定平台，其中的基础设施涉及住宅、商务和大批顾客。此外，Comcast 具有三个分区运营中心，它们管理本地基础设施以支持其所有顾客。国内和分区运营中心都通过利用公共进程和系统，以 $7 \times 24 \times 365$ 方式对其网络和产品性能的所有方面进行前摄性监视。例如，在国家和本地层面的各种网络事件具有共同预定义的安全性等级、恢复过程和达到预期恢复目标的平均时间。如果一个本地问题影响一个站点的运行，国内和分区运营中心能够相互支持。此外，国家和分区运营中心具有一个大规模的虚拟专用网，该虚拟专用网允许工程师安全地接入网络，从远地执行前摄性或反应性的网络管理活动。



这些屏幕显示了支持关联、阈值管理和由 Comcast 技术人员所使用的标签

Comcast 对网络管理的方法涉及 5 个关键区域：性能管理、故障管理、配置管理、账户管理和安全管理。性能管理（performance management）关注于理解以下方面：相对于每天的某时间、每周的某天或特殊事件（例如，拳击赛引起的流量高潮或支付事件），网络/系统和应用程序（统称为生态系统）是如何执行的。这些预定义的性能测度遍及服务路径，从顾客的住宅或商务到整个网络，再到合伙人和对等方的接口点。此外，为确保该生态系统的正常运行，持续不断地运行综合事务。故障管理（fault management）定义为检测、做日志和理解可能影响顾客的异常的能力。Comcast 利用相关性引擎来适当地确定某事件的严重性并采取适当的动作，在它影响顾客之前消除或纠正潜在的问题。配置管理（configuration management）确保该生态系统的所有元素都采用硬件和软件的适当版本。使这些元素维持在它们巅峰的“黄金级”水平，有助于它们避免无意识的后果。账户管理（accounting management）确保运营中心明白无误地了解该生态系统的供给与利用。为确保运营中心在所有时间都具有有效地重路由流量的能力，这种功能特别重要。安全管理（security management）确保存在适当的控制，确保该生态系统得到有效的保护，以免受到不适当的访问。

网络运营中心及其所支持的生态系统并非是静态的。工程和运营人员要不断地重新评价预定义的性能测度和工具，以确保满足顾客对良好运行的期望。

9.3 因特网标准管理框架

与 SNMP 名字所暗示的相反，在因特网中进行网络管理远比一个在管理实体和它的代理之间传递管理数据的协议所涉及的内容要多得多，它的不断发展已经比“简单”一词所暗示的要复杂得多。当前的因特网标准管理框架可追溯到简单网关监视协议——SGMP [RFC 1028]。SGMP 是由一批大学的网络研究者、用户和管理者所设计的，他们在 SGMP 方面的经验使他们仅在数月中设计、实现和部署了 SNMP [Lynch 1993]，这对于今天的延续时间相当漫长的标准化过程是一种触动。从那时起，SNMP 从 SNMPv1 到 SNMPv2，再演化到最近的版本 SNMPv3 [RFC 3410]，该版本于 1999 年 4 月公布，并于 2002 年 12 月更新。

当描述任何一种网络管理的框架时，不可避免地要涉及以下问题：

- 要监视什么（从语义的角度）？网络管理员能够执行什么形式的控制？
- 报告和/或交换的信息采用什么样的格式？
- 交换这些信息采用什么样的通信协议？

回想前一节中我们人类组织的类比。老板和分支机构经理将需要认可用来报告分支机构状态的活动、生产率和预算的度量方法。类似地，他们将需要认可老板能够采取的措施（例如，削减预算，命令分支机构经理改变该办事处运作的某些方面，或解雇职员和关闭分支机构）。在较低层次的细节方面，他们将需要认可数据报告的格式。例如，报告预算时采用何种货币（美元或欧元）？生产率将以什么单位来度量？虽然这些可能看起来是微不足道的细节，但无论如何它们必须要达成一致。最后，必须规定信息在总公司办事处和分支机构之间传递的方式（即它们的通信协议）。

因特网标准的管理框架可以解决上面提出的问题。该框架由 4 个部分组成：

- 网络管理对象的定义。这些对象被称为 MIB 对象。在因特网标准管理框架中，管理信息表现为管理对象的集合，这些对象共同形成了一个虚拟信息存储，称之为管理信息库（Management Information Base, MIB）。一个 MIB 对象可以是一个计数器，例如：因 IP 数据报首部差错而导致路由器丢弃的 IP 数据报的数量，或在以太网接口卡载波侦听到的差错数量；运行在一台 DNS 服务器上的软件版本的描述性信息；一个特定设备功能是否正确的状态信息；或到一个目的地的路由选择路径这种协议相关的信息等。MIB 对象因此定义了由被管设备维护的管理信息。相关的 MIB 对象聚集在 MIB 模块（module）中。在我们人类组织的类比中，该 MIB 定义了分支机构和总公司办事处之间传递的信息。
- 数据定义语言。该语言被称为管理信息结构（SMI）。SMI 定义了数据类型、对象模型，以及写入和修改管理信息的规则。MIB 对象用这种数据定义语言进行定义。在我们人类组织的类比中，SMI 用于定义被交换的信息格式的细节。
- 协议 SNMP。它用于在管理实体和代理之间传递信息和命令，代理代表在被管网络设备中的实体而执行操作。
- 安全性和管理能力。这些能力的增加代表了 SNMPv3 在 SNMPv2 基础上的加强。

因特网管理体系结构采用模块化的设计，采用了一种协议无关的数据定义语言和 MIB，以及一种 MIB 无关的协议。有意思的是，这种模块化的体系结构最先提出的原因是，使基于 SNMP 的网络管理易于向由国际标准化组织研制的网络管理框架迁移。后者是在最初构想 SNMP 时所提出的一种竞争性的网络管理体系结构。然而，这种迁移并没有出现。随着时间推移，SNMP 的设计方法使它进行了 3 个主要版本的演化，并且允许上面讨论的 SNMP 的 4 个主要部分独立演化。显然，过去所作的模块化决定是正确的，即使其原因是错误的！

在下面的内容中，我们将更为详细地学习因特网标准管理框架的这 4 个主要组件。

9.3.1 管理信息结构：SMI

管理信息结构（Structure of Management Information, SMI）是一个名字相当奇特的网络管理框架的组件，从它的名字看不出它的功能。SMI 是用于定义驻留在被管网络实体中的管理信息的语言。为确保网络管理数据的语法和语义定义明确和无二义性，需要这样一种定义语言。注意到 SMI 不是用于定义被管网络实体中特定数据的实例的，而是定义这种信息的语言。用于描述 SNMPv3 的 SMI 文档（它被称为 SMIV2，这特别容易让人混淆）是 [RFC 2578; RFC 2579; RFC 2580]。我们以自底向上的方式来研究 SMI，并从 SMI 中的基本数据类型开始。然后我们观察如何用 SMI 来描述被管对象，以及相关的被管对象是如何分组而成为模块的。

1. SMI 基本数据类型

RFC 2578 规定了用于 SMI MIB 模块定义语言中的基本数据类型。尽管 SMI 基于 ASN.1（抽象语法记法 1）[ISO X.680 2002] 对象定义语言（参见 9.4 节），但它已经增加了足够多的 SMI 特定数据类型，因而可以认为 SMI 是具有专门用途的数据定义语言。在 RFC 2578 中定义的 11 种基本数据类型显示在表 9-1 中。除了这些标量对象外，还可以使用 SEQUENCE OF 结构也可能将一些有序的 MIB 对象集合组合成列结构；细节请参见 RFC 2578。对多数读者而言，表 9-1 中的多数数据类型将是熟悉的（或自解释的）。我们很快将更详细地讨论一种数据类型，即 OBJECT IDENTIFIER 的数据类型，它被用于命名对象。

表 9-1 SMI 的基本数据类型

数据类型	描述
INTEGER	32 比特的整数，如 ASN.1 所定义，其值在 -2^{31} 和 $2^{31} - 1$ 之间（含端值），或一个来自可能的命名常数值列表的值
Integer32	32 比特的整数，其值在 -2^{31} 和 $2^{31} - 1$ 之间（含端值）
Unsigned32	无符号的整数，其值在 0 和 $2^{32} - 1$ 之间（含端值）
OCTET STRING	ASN.1 格式字节串，表示任意二进制或文本数据，长度最多为 65 535 字节
OBJECT IDENTIFIER	ANS.1 格式，由管理指派（结构化名称），参见 9.3.2 节
IPAddress	32 比特因特网地址，以网络字节顺序
Counter32	32 比特计数器，能从 0 到 $2^{32} - 1$ 增加，然后回归到 0
Counter64	64 比特计数器
Gauge32	32 比特整数，当增加或减少时，它不能增加到 $2^{32} - 1$ 以上，也不能降到 0 以下
TimeTicks	时间，自某事件起计算，精度 0.01 秒
Opaque	未解释的 ASN.1 字符串，用于向后兼容

2. SMI 较高层结构

除了基本数据类型外，SMI 数据定义语言也提供了较高层的语言结构。

OBJECT-TYPE 结构用于定义被管对象的数据类型、状态和语义。概括来说，这些数据对象包含了位于网络管理内部的管理数据。在各种因特网 RFC 中有大约 10 000 个定义的对象 [RFC 3410]。OBJECT-TYPE 结构具有 4 条子句。OBJECT-TYPE 定义的 SYNTAX 子句规定了与对象相关的基本数据类型。MAX-ACCESS 子句规定了被管对象是否能读、能写、能创建，或在一个通知中包括有它的值。STATUS 子句指出了该对象定义是否是当前的和合法的、过时的（在这种情况下，它不应当被实现，因为该定义仅为了历史原因而出现），或不赞成的（过时的，但为了与过去的实现互操作而实现）。DESCRIPTION 子句包含有关该对象的人可读的文本定义；它是被管对象用途的文档，提供了实现该被管对象所需的所有语义信息。

作为 OBJECT-TYPE 结构的一个例子，考虑来自 [RFC 4293] 的 ipSystemStatsInDelivers 对象类型定义。该对象定义了一个 32 比特的计数器，用以跟踪被管设备接收到并成功地传递给较高层协议的 IP 数据报的数量。该定义的最后一行与这个对象的名字有关，这是我们将在下一节中要考虑的主题。

```
ipSystemStatsInDelivers OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The total number of datagrams successfully
        delivered to IPuser-protocols (including ICMP).

        When tracking interface statistics, the counter
        of the interface to which these datagrams were
        addressed is incremented. This interface might
        not be the same as the input interface for
        some of the datagrams.

        Discontinuities in the value of this counter can
        occur at re-initialization of the management
        system, and at other times as indicated by the
        value of ipSystemStatsDiscontinuityTime."
    ::= { ipSystemStatsEntry 18 }
```

MODULE-IDENTITY 结构允许相关的对象分组在一个“模块”中。例如，[RFC 4293] 规定了用于定义被管对象（包括 ipSystemStatsInDelivers）的 MIB 模块，这些被管对象用于网

际协议（IP）和与之相关的因特网控制报文协议（ICMP）的管理实现。[RFC 4022] 规定了用于 TCP 的 MIB 模块，[RFC 4113] 规定了用于 UDP 的 MIB 模块。[RFC 4502] 定义了用于 RMON 远程监视的 MIB 模块。除了包括该模块中的被管对象的 OBJECT-TYPE 定义，MODULE-IDENTITY 结构还包含了该模块作者的联系信息、最后更新的日期、修订历史和该模块的文本性描述。举一个例子，考虑下面用于 IP 协议管理的模块定义的例子：

```
ipMIB MODULE-IDENTITY
    LAST-UPDATED "200602020000Z"
    ORGANIZATION "IETF IPv6 MIB Revision Team"
    CONTACT-INFO
        "Editor:
        Shawn A. Routhier
        Interworking Labs
        108 Whispering Pines Dr. Suite 235
        Scotts Valley, CA 95066
        USA
        EMail: <sar@iwl.com>"
    DESCRIPTION
        "The MIB module for managing IP and ICMP
        implementations, but excluding their
        management of IP routes.

        Copyright (C) The Internet Society (2006).
        This version of this MIB module is part of
        RFC 4293; see the RFC itself for full legal
        notices."

    REVISION      "200602020000Z"
    DESCRIPTION
        "The IP version neutral revision with added
        IPv6 objects for ND, default routers, and
        router advertisements. As well as being the
        successor to RFC 2011, this MIB is also the
        successor to RFCs 2465 and 2466. Published
        as RFC 4293."

    REVISION      "199411010000Z"
    DESCRIPTION
        "A separate MIB module (IP-MIB) for IP and
        ICMP management objects. Published as RFC
        2011."

    REVISION      "199103310000Z"
    DESCRIPTION
        "The initial revision of this MIB module was
        part of MIB-II, which was published as RFC
        1213."
::= { mib-2 48}
```

NOTIFICATION-TYPE 结构用来定义由代理或管理实体产生的有关“SNMPv2-Trap”和“InformationRequest”报文的信息，参见 9.3.3 节。该信息包括了文本性 DESCRIPTION：何时发送这样的报文，以及包括在产生的报文中的值列表；详情参见 [RFC 2578]。MODULE-COMPLIANCE 结构定义了一个代理必须实现的模块中的被管对象集合。AGENT-CAPABILITIES 结构定义了代理关于对象通知和事件通知定义的能力。

9.3.2 管理信息库：MIB

如上所述，管理信息库（Management Information Base，MIB）能被认为是一个虚拟信

息库，库中的被管对象的值总体上反映了该网络的当前“状态”。查询和/或设置这些值，可以由管理实体通过向代理发送 SNMP 报文来进行，而代理代表管理实体并在被管设备上执行操作。被管对象使用前面讨论的 OBJECT-TYPE SMI 结构来定义，并使用 MODULE-IDENTITY 结构汇集在 MIB 模块（MIB Module）中。

IETF 已经专注于对与路由器、主机和其他网络设备相关的 MIB 模块进行标准化。这包括了有关硬件的特定部分的基本标识数据及有关设备网络接口和协议的管理信息。到 2006 年，有 200 多个基于标准的 MIB 模块，以及更多的厂商特定（专用）的 MIB 模块。对于所有这些标准，IETF 需要一种方法来标识和命名标准化的模块以及模块中的特定被管对象。IETF 没有白手起家，而是采用了国际标准化组织（ISO）已经提出的一种标准化对象标识（命名）框架。由于具有许多标准化组织的缘故，ISO 对于它们的标准化对象标识框架有“庞大计划”，即标识任何网络中的每个可能的标准化对象（如数据格式、协议或部分信息），而无论该对象是由哪个网络标准组织（例如，因特网 IETF、ISO、IEEE 或 ANSI）、设备制造商或网络所有者所定义的。这的确是一个极高的目标！由 ISO 采纳的对象标识框架是 ASN.1（抽象语法记法 1）对象定义语言的一部分 [ISO X.680 2002]，我们将在 9.4 节中讨论 ASN.1。标准的 MIB 模块在这个无所不包的命名框架中有它们自己的合适位置，如下面所讨论。

如图 9-3 所示，对象在 ISO 命名框架中以等级结构方式进行命名。注意到在该树上的每个分支点具有一个名字和一个编码（显示在圆括号中）；因而该树中的任何点可由名字或编码的序列所标识，它规定了从根到标识树的那个点的路径。一个有趣的（但不完整和非正式的）基于 Web 的实用程序能用于浏览部分该对象标识树（使用志愿者提供的分支信息），有关信息能在 [OID Repository 2012] 中找到。

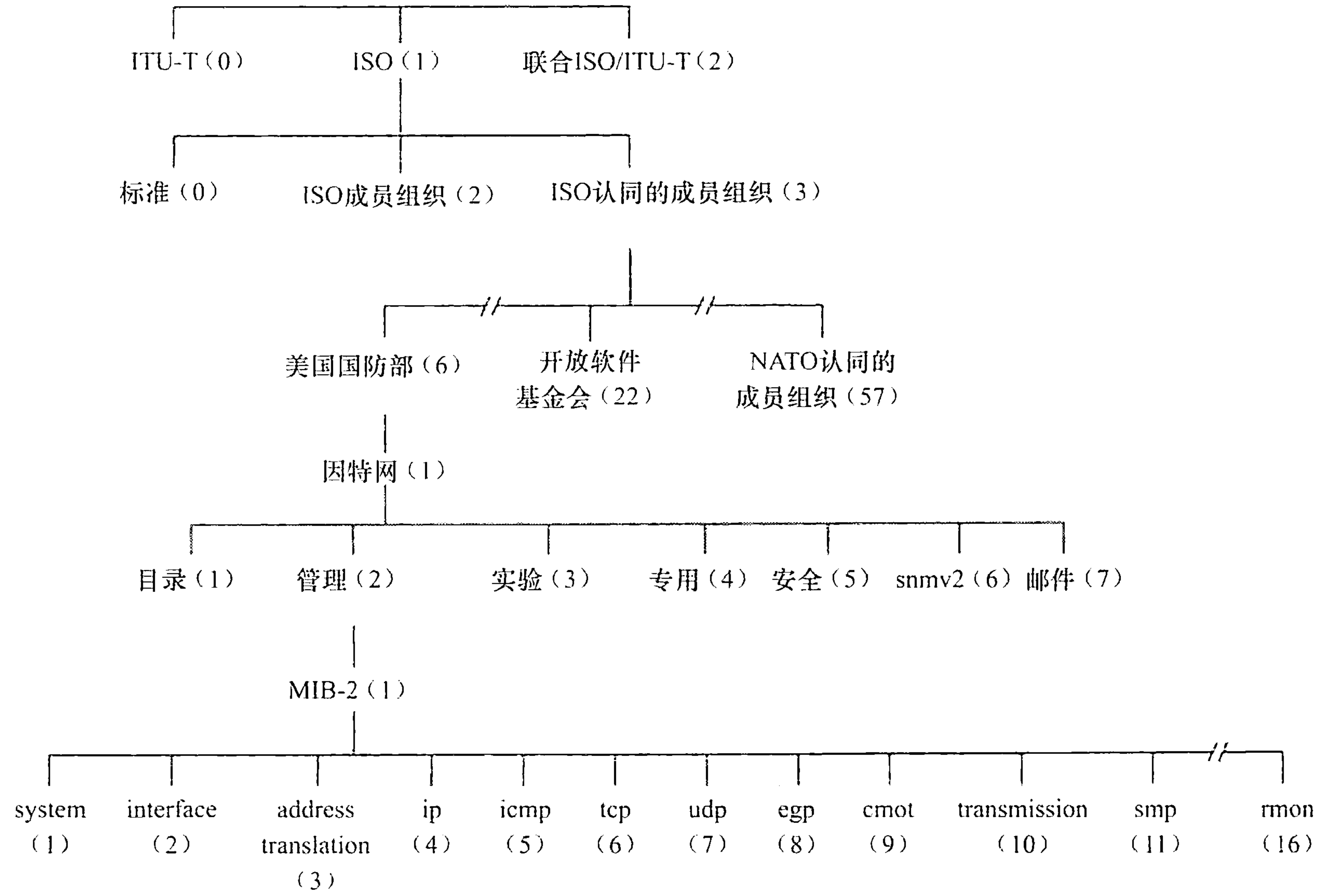


图 9-3 ANS. 1 对象标识树

在该等级结构顶部是 ISO 和国际电信联盟的电信标准化组织（ITU-T），以及这两个组织联合工作的一个分支机构。这两个主要的标准化组织共同研究 ANS. 1。在这棵树的 ISO 分支下，我们还发现一些条目，它们分别对应所有 ISO 标准（1.0）和各个 ISO 成员国的标准组织所发布的标准（1.2）。尽管没有在图 9-3 中显示出来，但在 ISO 成员组织（又可表示为 1.2）下，我们能够发现 USA（1.2.840），在它下面我们将能发现 IEEE、ANSI 和公司特定标准的编码，其中包括 RSA（1.2.840.11359）和微软（1.2.840.113556）。在微软下面，我们能发现用于各种微软产品的微软文件格式（1.2.840.113556.4），如 Word（1.2.840.113556.4.2）。但是我们现在感兴趣的是网络（并非微软的 Word 文件），因此我们将注意力转向标号为 1.3 的分支，这些标准由 ISO 认可的组织发布。这些组织包括美国国防部（6）（在它下面我们能发现因特网标准）、开放软件基金会（Open Software Foundation）（22）、航空协会 SITA（69）和北大西洋公约组织（NATO）认可的成员组织（57），以及许多其他组织。

在该树的因特网分支（1.3.6.1）下面，有 7 个类别。在 private（1.3.6.1.4）分支下，列出了名字和专用企业编码的列表 [IANA 2009b]，该列表有超过几千个已经在因特网编号分配机构（Internet Assigned Number Authority, IANA）注册的专门公司 [IANA 2009a]。在对象标识树的 management（1.3.6.1.2）和 MIB-2（1.3.6.1.2.1）分支下，我们发现标准 MIB 模块的定义。到达我们在 ISO 名字空间的角落需要经历多么漫长的旅行啊！

标准化的 MIB 模块

在图 9-3 中树的最底层显示了某些重要的面向硬件的 MIB 模块（system 和 interface）以及与某些最重要的因特网协议相关的模块。[RFC 5000] 列出了自 2008 年以来所有的标准 MIB 模块。虽然阅读 MIB 相关的 RFC 相当乏味和枯燥，但考虑一些 MIB 模块定义，对认识模块中的信息类型是有指导意义的（就像吃蔬菜对你身体有好处一样）。

位于 system 之下的被管对象包含了有关被管设备的一般性信息；所有被管设备必须支持 system MIB 对象。表 9-2 定义了 system 组中的对象，这些对象由 [RFC 1213] 所定义。表 9-3 定义了一个被管实体中用于 UDP 协议的 MIB 模块中的被管对象。

表 9-2 在 MIB-2 system 组中的被管对象

对象标识符	名字	类型	描述（引自 RFC 1213）
1.3.6.1.2.1.1.1	sysDescr	OCTET STRING	“该系统的硬件类型、软件操作系统和网络软件的全名和版本标识”
1.3.6.1.2.1.1.2	sysObjectID	OCTET IDENTIFIER	分配给厂商的对象 ID，“它提供了一种易于操作和无二义性的方法以决定被管理的‘单元的类型’”
1.3.6.1.2.1.1.3	sysUpTime	TimeTicks	“自系统的网络管理部分最后被重新初始化以来的时间（精度为 0.01 秒）”
1.3.6.1.2.1.1.4	sysContact	OCTET STRING	“该被管结点的联系人，以及关于该人联系方式的信息”
1.3.6.1.2.1.1.5	sysName	OCTET STRING	“为该结点正式分配的名字。按惯例，这是该结点的全称域名”
1.3.6.1.2.1.1.6	sysLocation	OCTET STRING	“该结点的物理位置”
1.3.6.1.2.1.1.7	sysServices	Integer32	“指出在该结点可用的服务集合的编码值：物理的（如一个转发器），数据链路/子网（如网桥），因特网（如 IP 网关），端到端（如主机），应用程序

表 9-3 在 MIB-2 UDP 模块中的部分被管对象

对象标识符	名字	类型	描述（引自 RFC 4113）
1.3.6.1.2.1.7.1	udpInDatagrams	Counter32	“交付给 UDP 用户的 UDP 数据报总数”
1.3.6.1.2.1.7.2	udpNoPorts	Counter32	“在那些没有应用程序的目的地端口所接收到的 UDP 数据报总数”
1.3.6.1.2.1.7.3	udpInErrors	Counter32	“接收的不能传递的 UDP 数据报数量，其原因并非 是目的地端口没有应用程序”
1.3.6.1.2.1.7.4	udpOutDatagrams	Counter32	“从该实体发送的 UDP 数据报总数”

9.3.3 SNMP 协议运行和传输映射

简单网络管理协议版本 2（SNMPv2）[RFC 3416] 用于管理实体和代表管理实体执行的代理之间传递 MIB 信息。SNMP 的最常使用方式是请求响应模式（request-response mode），其中 SNMPv2 管理实体向 SNMPv2 代理发送一个请求，代理接收到该请求后，执行某些动作，然后向该请求发送一个回答。请求通常用于查询（检索）或修改（设置）与某被管设备相关的 MIB 对象值。第二个常使用的 SNMP 报文是代理向管理实体发送的一种非请求报文，该报文称为陷阱报文（trap message）。该报文用于通知管理实体，一个异常情况已经导致了 MIB 对象值的改变。我们在 9.1 节中已看到，网络管理员在出现如下情况时可能希望接收到陷阱报文，例如，一个接口不工作，某链路上的拥塞达到一个预定的级别，或发生了某些其他值得注意的事件。注意到在轮询（请求响应交互）和陷阱之间有许多折中；参见课后习题。

表 9-4 中表示了 SNMPv2 定义的 7 种类型的报文，这些报文一般称为协议数据单元（PDU）。图 9-4 显示了这些 PDU 的格式。

表 9-4 SNMPv2 PDU 类型

SNMPv2 PDU 类型	发送方 - 接收方	描述
GetRequest	管理者到代理	取得一个或多个 MIB 对象实例值
GetNextRequest	管理者到代理	取得列表中下一个 MIB 对象实例值
GetBulkRequest	管理者到代理	以大数据块方式取得值，例如大表中的值
InformRequest	管理者到管理者	向不能访问的远程管理实体通知 MIB 值
SetRequest	管理者到代理	设置一个或多个 MIB 对象实例的值
Response	代理到管理者或管理者到管理者	对 GetRequest、GetNextRequest、GetBulkRequest、SetRequest PDU 或 InformRequest 产生的响应
SNMPv2-Trap	代理到管理者	向管理者通知一个异常事件

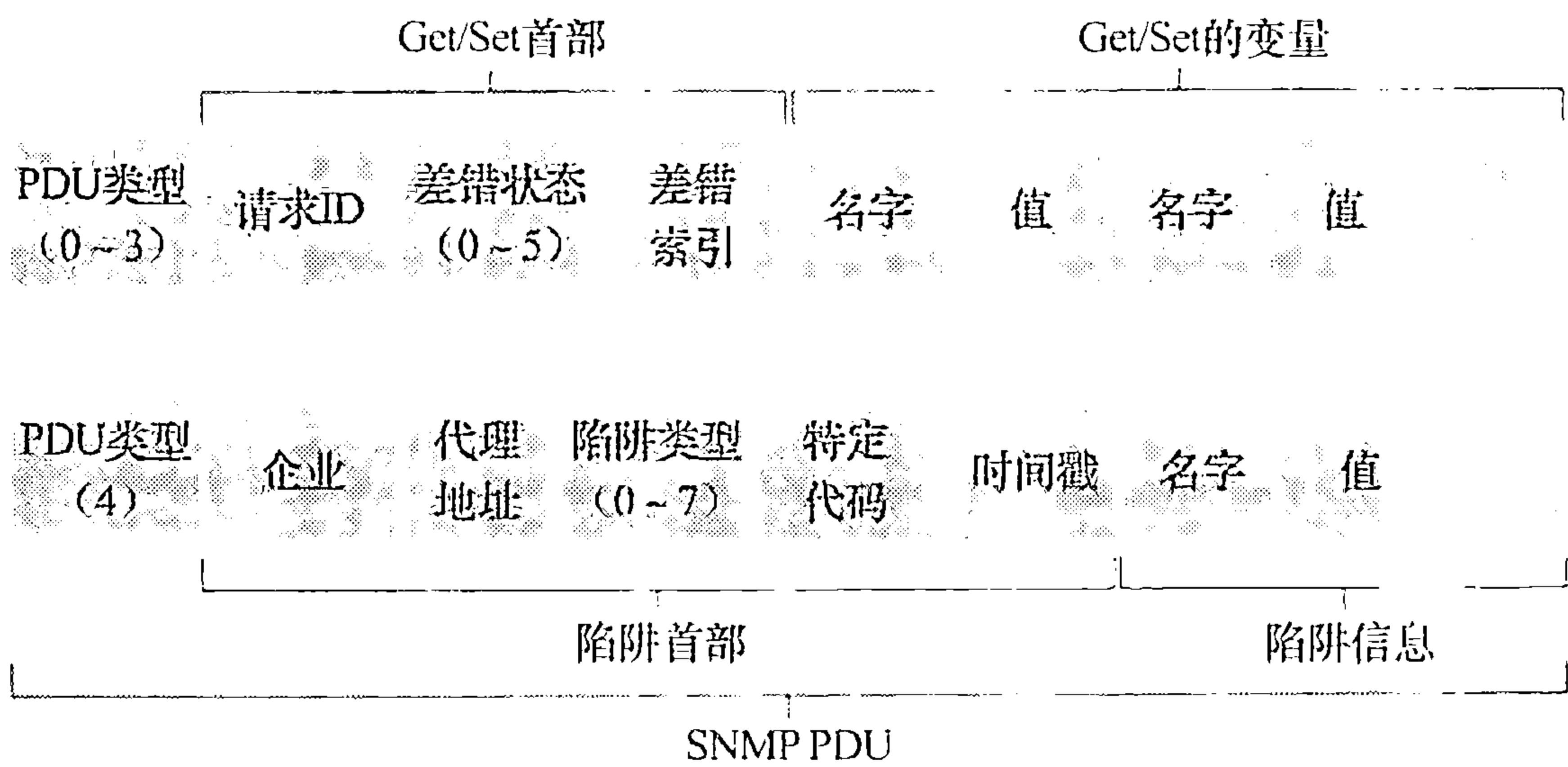


图 9-4 SNMP PDU 格式

- GetRequest、GetNextRequest 和 GetBulkRequest PDU 都是管理实体向代理发送的，以请求位于该代理所在的被管设备中的一个或多个 MIB 对象值。其值被请求的 MIB 对象的对象标识符定义在该 PDU 的变量绑定部分。GetRequest、GetNextRequest 和 GetBulkRequest 的差异在于它们的数据请求粒度。GetRequest 能够请求 MIB 值的任意集合；多个 GetNextRequest 能用于顺序地读取 MIB 对象的序列或表；GetBulkRequest 允许读取大块数据，能够避免因发送多个 GetRequest 或 GetNextRequest 报文可能导致的额外开销。在所有这三种情况下，代理用包括该对象标识符和它们相关值的 Response PDU 进行响应。
- SetRequest PDU 是管理实体用来设置在被管设备中的一个或多个 MIB 对象值的。代理用带有“无错（noError）”差错状态的 Response PDU 来回答，证实该值的确已经被设置。
- InformRequest PDU 是一个管理实体用来通知另一个管理实体的，其中 MIB 信息是接收实体不能访问的。该接收实体用带有“无错（noError）”差错状态的 Response PDU 来确认收到了 InformRequest PDU。
- SNMPv2 PDU 的最后一种类型是陷阱报文。陷阱报文是异步产生的，即它们不是为了响应接收到的请求而产生的，而是为了响应管理实体所需要的通知事件而产生的。RFC 3418 定义了周知的陷阱类型，其中包括设备进行的冷启动或热启动，链路的从故障状态变为工作状态或相反，找不到相邻设备，或鉴别失效事件。接收到的陷阱请求不要求从管理实体得到响应。

给出了 SNMPv2 请求响应性质后，注意到尽管 SNMP PDU 能够通过许多不同的运输协议传输，但 SNMP PDU 通常还是在 UDP 数据报的负载中传输的。事实上，RFC 3417 认为 UDP 是“首选的运输映射”。由于 UDP 是一种不可靠的运输协议，因而不能确保一个请求或它的响应将能够被它的目的地接收到。管理实体用该 PDU 的“请求 ID”字段为它向代理发送的请求编号；代理的响应从接收到的请求中获取该“请求 ID”。因此，该“请求 ID”字段能被管理实体用来监测丢失的请求或回答。如果在一定量时间后还没有收到对应的响应，由管理实体来决定是否重传一个请求。特别是，SNMP 标准没有强制任何特殊的重传过程，哪怕首先进行了重传。它只是要求管理实体“需要根据重传的频率和周期可靠地动作”。这当然让人们想知道一个“可靠的”协议行为应当如何做！

9.3.4 安全性和管理

SNMPv3 的设计者们说过“SNMPv3 能被看成是具有附加的安全性和管理能力的 SNMPv2” [RFC 3410]。SNMPv3 比起 SNMPv2 来无疑有一些变化，但是其他地方的变化都没有管理和安全性领域的变化那么明显。在 SNMPv3 中安全性的重要性尤为突出，因为缺乏安全性导致 SNMP 只能用于监视而不能用于控制（例如，在 SNMPv1 中很少使用 SetRequest）。

SNMP 经历了 3 个版本后已经成熟，它的功能有了增加，但不幸的是，与 SNMP 相关的标准文档的数量增加了。这能够由下述事实来印证：现在甚至有了“描述用于描述 SNMP 管理框架的体系结构的 RFC” [RFC 3411]！虽然描述一个框架的体系结构这个概念也许使人有些不太好理解，但 RFC 3411 的目的是值得肯定的，它引入了一种共同语言，用于描述功能和由 SNMPv3 代理或管理实体所采取的动作。SNMPv3 实体的体系结构是简单明了的，浏览该体系结构将有助于我们进一步理解 SNMP。

所谓 SNMP 应用程序 (SNMP application) 是由命令产生器、通知接收器和代理转发器 (这些通常位于管理实体中)、命令响应器和通知源发器 (这两者通常位于代理中) 以及其他可能的应用程序组成的。命令产生器产生我们在 9.3.3 节讨论过的 GetRequest、GetNextRequest、GetBulkRequest 和 SetRequest PDU, 并处理对这些 PDU 接收到的响应。命令响应器在代理中执行, 接收、处理和 (用 Response 报文) 回答接收到的 GetRequest、GetNextRequest、GetBulkRequest 和 SetRequest PDU。代理中的通知源发器应用程序产生 Trap PDU; 这些 PDU 最终由管理实体中的通知接收器应用程序接收和处理。代理转发器应用程序转发请求 PDU、通知 PDU 和响应 PDU。

由 SNMP 应用程序发送的 PDU 在经过适当的运输协议发送之前, 接下来通过 SNMP “引擎” 传递。图 9-5 显示了由命令产生器应用程序产生的一个 PDU 先进入调度模块, 在那里决定 SNMP 的版本。然后该 PDU 在报文处理系统中进行处理, 在这里该 PDU 被封装在包括 SNMP 版本号、报文 ID 和报文长度信息的报文首部中。如果需要加密或鉴别, 还要包括针对这些信息的适当首部字段信息; 详见 [RFC 3411]。最后, SNMP 报文 (应用程序产生的 PDU 加上首部信息的报文) 被传递到适当的运输协议。携带 SNMP 报文的首选运输协议是 UDP (即 SNMP 报文被作为 UDP 数据报的负载传送), 用于 SNMP 的首选端口号是端口 161。端口 162 用于陷阱报文。

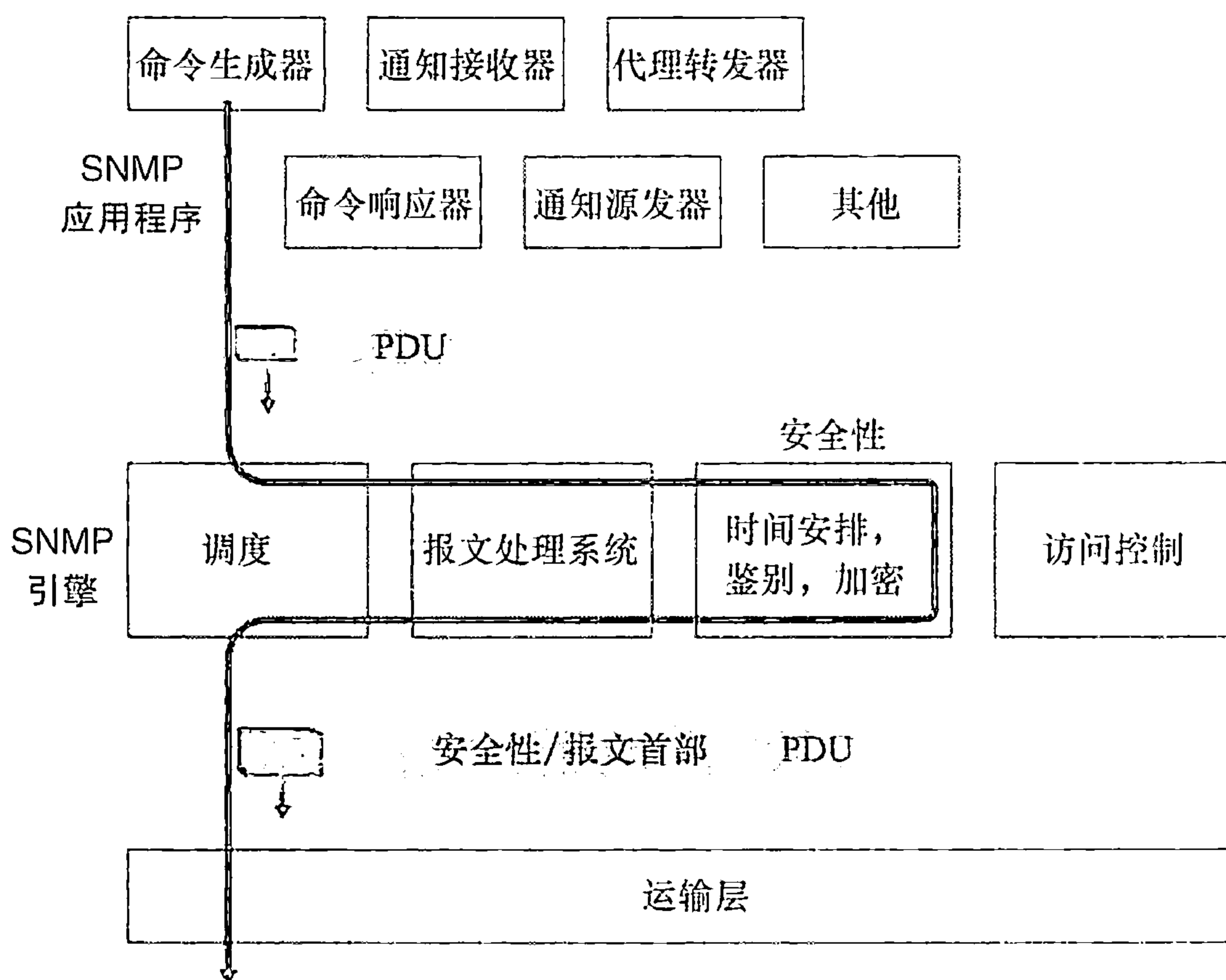


图 9-5 SNMPv3 引擎和应用程序

前面我们已经看到 SNMP 报文不仅能用于监视, 也能用于控制 (例如, 通过 SetRequest 命令) 网络元素。显然, 若一个入侵者能够截获 SNMP 消息和/或产生它自己的 SNMP 报文并向管理基础设施发送, 它就可能会对网络造成损害。因此, 安全地传输 SNMP 报文是至关重要的。令人惊奇的是, 仅在 SNMP 最近的版本中, 安全性才得到应有的注意。SNMPv3 的安全性被称为基于用户的安全性 (user-based security) [RFC 3414], 这是因为它应用了用户的传统概念 (用户采用用户名来标识), 还有相关的口令、密码值或访问权限等安全信息。SNMPv3 提供了加密、鉴别、对重放攻击的防护 (参见 8.3 节) 和访问控制功能。

- 加密。SNMP PDU 能够使用密码分组链接模式的数据加密标准 (DES) 进行加密。注意到因为 DES 是一个共享密钥的系统, 故用户用于加密数据的密钥必须被要对该数据解密的接收实体所知道。
- 鉴别。SNMP 使用了报文鉴别码 (MAC) 技术, 以提供鉴别并保护数据不受篡改 [RFC 4301], 其中的 MAC 我们已在 8.3.1 节中学习过。前面讲过 MAC 要求发送方和接收方都知道一个共同的密钥。
- 对重放攻击的防护。在第 8 章中我们讨论了不重数能被用于防护重放攻击。SNMPv3 采用了相关方法。为了确保一个接收的报文不是某个先前报文的重复, 该接收方要求发送方在每个报文中包括一个基于接收方中的计数器的值。该计数器的功能与不重数相同, 反映了自接收方的网络管理软件最后一次启动以来的时间和自接收方的网络管理软件最后一次配置以来启动的总数。只要在接收报文中的计数器位于接收方的实际值的某种错误容限内, 该报文被认为是非重放攻击的报文, 此时它已被鉴别和/或加密。详见 [RFC 3414]。
- 访问控制。SNMPv3 提供了一种基于视图的访问控制 [RFC 3415], 该方法控制了哪些网络管理信息能被哪些用户查询和/或设置。一个 SNMP 实体在本地配置库 (Local Configuration Datastore, LCD) 保留了有关访问权限和策略的信息。LCD 的某些部分自身可作为被管对象访问, 在基于视图访问控制模型配置 MIB 中定义 [RFC 3415], 并因此能够经 SNMP 从远程进行管理和操作。

9.4 ASN.1

在本书中, 我们已经涉及了一些计算机网络的有趣主题。然而, 本节有关 ASN.1 的内容或许算不上前 10 个有趣的主题。像蔬菜有益于健康一样, 有关 ASN.1 的知识和表示服务的更广泛问题是对你有益的东西。ASN.1 是一种源于 ISO 的标准, 用于一些因特网相关的协议中, 特别是用于网络管理领域中。例如, 我们在 9.3 节中看到, SNMP 中的 MIB 变量与 ASN.1 有不解之缘。尽管 ASN.1 的有关资料可能相当枯燥, 但我们希望读者能够认识到这些相关材料的重要性。

为了激发我们下面的讨论, 这里考虑下列可能的实验。假定一个人能够可靠地将数据从一台计算机的内存直接复制到远程另一台计算机的内存中。如果他能够这样做, 其中的通信问题将是如何解决的? 该问题的答案取决于他对“通信问题”的定义。毫无疑问, 一个完善的内存到内存复制将是 从一台机器到另一台机器的比特和字节的精确通信。当在接收计算机上运行的软件存取这些数据时, 这些比特和字节的实际复制意味着什么? 我们能够看到与在发送计算机内存中所存储的一样的值吗? 对该问题的答案是“不一定”! 该问题的症结在于, 不同的计算机体系结构、不同的编译程序具有不同的存储和表示数据的常规方法。如果数据要通信, 或在多台计算机中存储 (因为数据在各通信网络中), 显然必须要解决数据表示的问题。

作为该问题的一个例子, 考虑下面简单的 C 代码段。这个结构怎样在内存中安排呢?

```
struct {  
    char code;  
    int x;  
} test;  
test.x = 259;  
test.code = 'a';
```

图 9-6 的左侧显示了该数据在一种假想的体系结构中的一种可能的内存安排：在一个单字节内存中包含了字符“a”，其后的 16 比特的字包含了整数值 259，以高阶字节在前的顺序存储。在另一台计算机中的内存安排显示在图 9-6 的右侧。字符“a”后面是 16 比特的整数值，以低阶字节在前的顺序存储，它们以 16 比特的字为边界对齐。当然，如果某人在这两台机器的内存之间执行逐字复制，使用相同的结构定义来存取存储的值，他将在这两台计算机上看到极为不同的结果！

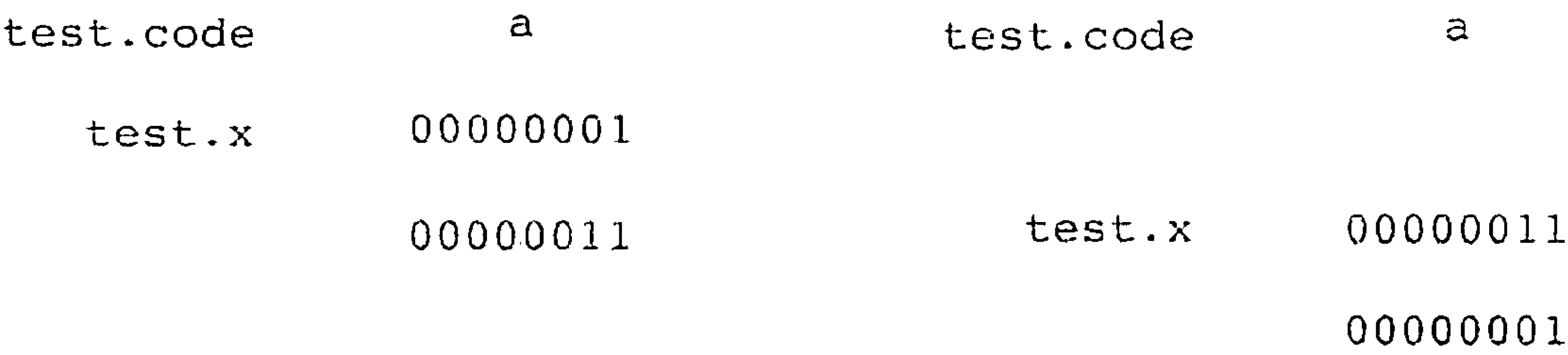


图 9-6 两种不同体系结构上的两种不同的数据安排

不同的体系结构具有不同的内部数据格式，这是一个真实而普遍的问题。以不同的形式存储整数的特定问题非常常见，并且有了一个专有名称。用于存储整数的“大端法”（big-endian）次序要求最高阶字节先存储（在最低的存储地址），“小端法”（little-endian）次序要求先存储最低阶字节。Sun SPARC 和 Motorola 处理器采用“大端法”，而 Intel 和 DEC/Compaq Alpha 处理器采用“小端法”。作为一个小插曲提一下，术语“big-endian”和“little-endian”来自 Jonathan Swift 的书《格列佛游记》，书中的两群人固执己见地坚持以两种不同方式做一件简单的事情（希望对计算机体系结构方面的这个类比有助于理解）。来自小人国的一群人坚持在较大一端打开鸡蛋（“大端法”），而其他人坚持在较小一端打开鸡蛋。这个分歧导致了大规模民事冲突和叛乱。

既然存在不同方式的计算机存储和数据表示，网络协议该怎样处理这些问题呢？例如，如果一个 SNMP 代理打算发送一个响应报文，其中包含接收到的 UDP 数据报数量的整数计数器的值，它怎样表示将向管理实体发送的该整数值呢？是以大端法还是以小端法次序呢？代理的一种选择是以在管理实体中存储的相同顺序来发送这些字节。另一种选择是代理以自己的存储顺序发送，并让接收实体根据需要重新排序。任何一种选择都将要求发送方或接收方知道其他人的整数表示格式。

第三种选择是用与机器无关、OS 无关、语言无关的方式描述整数和其他数据类型（即一种数据定义语言）和规则，该规则定义了每种数据类型跨越网络传输的方式。当接收到给定类型的数据时，它以一种已知格式来接收，并能以任意机器特定的格式进行存储。我们在 9.3 节中学习的 SMI 和 ASN.1 都采用了第三种选择。用 ISO 术语来说，这两个标准描述了一种表示服务（presentation service），该服务用于将信息从一种机器特定格式传输和转换到另一种机器特定格式。图 9-7 阐述了一个真实世界中的表示问题；两个接收方都不理解交流的基本思想，如谈话者喜欢的东西。如图 9-8 所示，一个表示服务能够解决该问题的过程是，通过将思想转换成共同理解的（通过表示服务）、与个人无关的语言，向接收方发送该信息，然后转换成接收方能理解的语言。

表 9-5 显示了一些 ASN.1 定义的数据类型。前面我们在学习 SMI 时已遇到了 INTEGER、OCTET STRING 和 OBJECT IDENTIFIER 数据类型。这里我们的目标不是提供对 ASN.1 的完整概述，因此我们向读者推荐描述 ASN.1 类型和结构的标准或已出版的和在线的书籍

[Larmouth 1996]。这些结构如 SEQUENCE 和 SET，它们可用于定义结构化的数据类型。

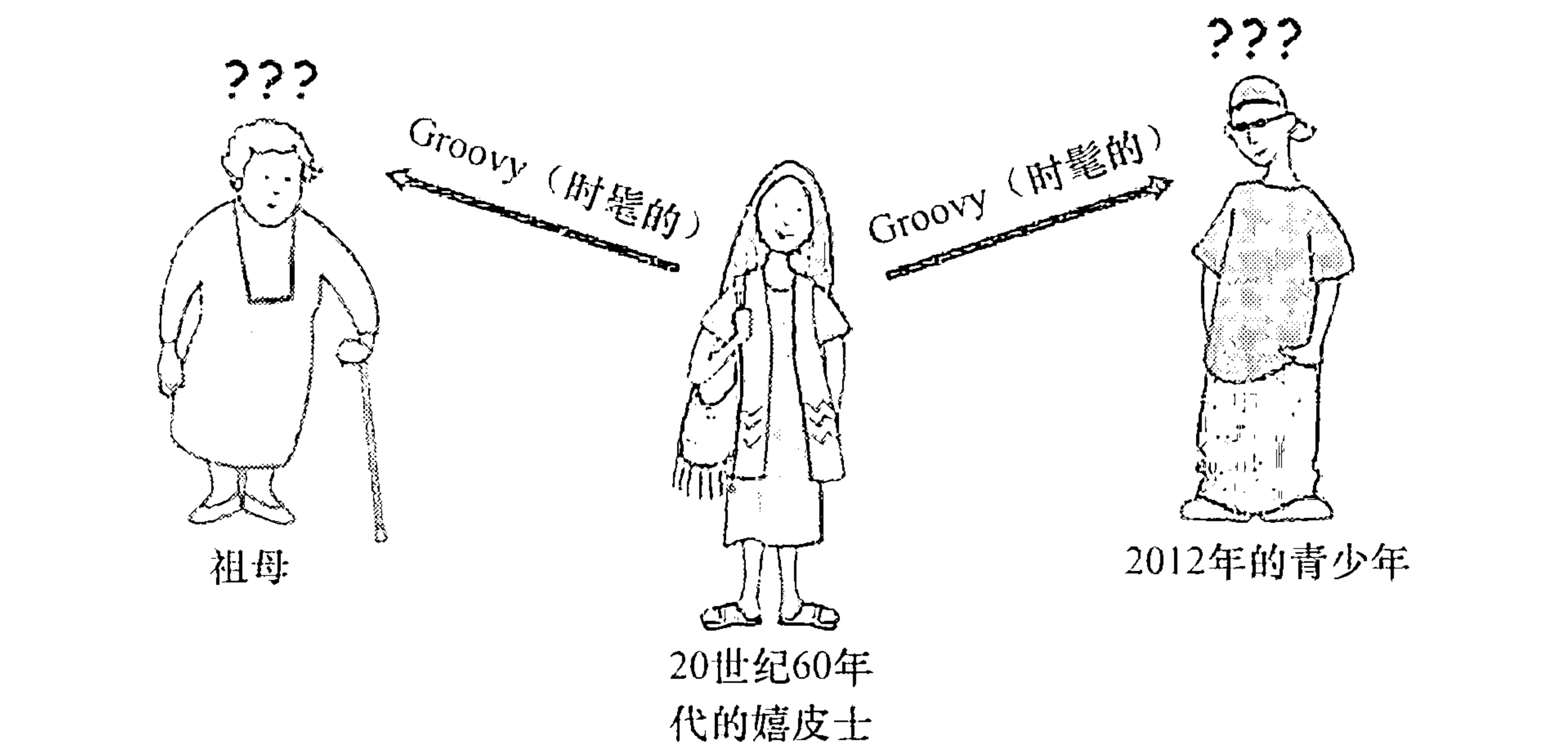


图 9-7 表示问题

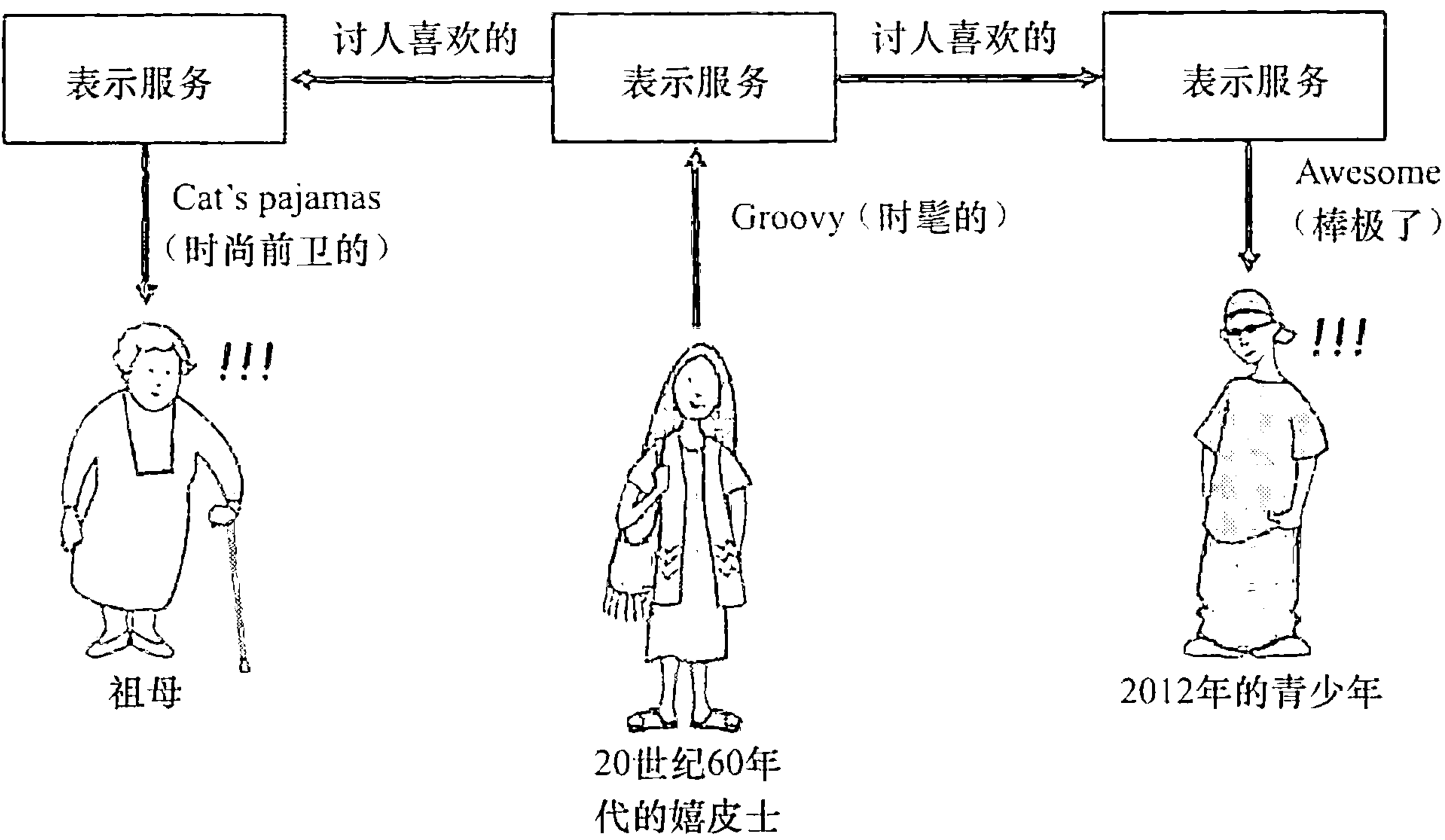


图 9-8 表示问题的解决

除了提供数据定义语言，ASN.1 也提供基本编码规则（Basic Encoding Rules，BER）。BER 定义了采用 ASN.1 数据定义语言所定义的对象实例经由网络发送的方式。BER 采用一个所谓 TLV（Type，Length，Value）方法（TLV approach）对发送的数据进行编码。对于被发送的每个数据项，将以该数据项的数据类型、长度以及它的实际值的顺序进行发送。对于这个简单的规则，接收的数据基本上是自标识的。

表 9-5 部分 ASN.1 数据类型

标志	类型	描述
1	BOOLEAN	值为“真”或“假”
2	INTEGER	能够为任意大
3	BIT STRING	一个或多个比特的列表
4	OCTET STRING	一个或多个字节的列表
5	NULL	无值
6	OBJECT IDENTIFIER	名字，位于 ASN.1 标准命名树上，参见 9.2.2 节
9	REAL	浮点

图 9-9 显示了如何发送两个数据项的一个简单例子。在这个例子中，假定采用大端法顺序，发送方要发送字符串“smith”后面带有值为 259 的十进制数（这等于二进制数 00000001 00000011，或值为 1 的一个字节后跟值为 3 的一个字节）。这个传输流中的第一个字节具有值 4，指出后面的数据项的类型是 OCTET STRING；这是 TLV 编码中的“T”。该流中的第二个字节包含了 OCTET STRING 的长度，本例中是 5。在该传输流中的第三个字节开始了长度为 5 的 OCTET STRING；它包含了字母“s”的 ASCII 表示。下个数据项的 T、L 和 V 的值是 2（INTEGER 类型标志值）、2（即长度为 2 字节的一个整数）和十进制数 259 的 2 字节的大端法表示。

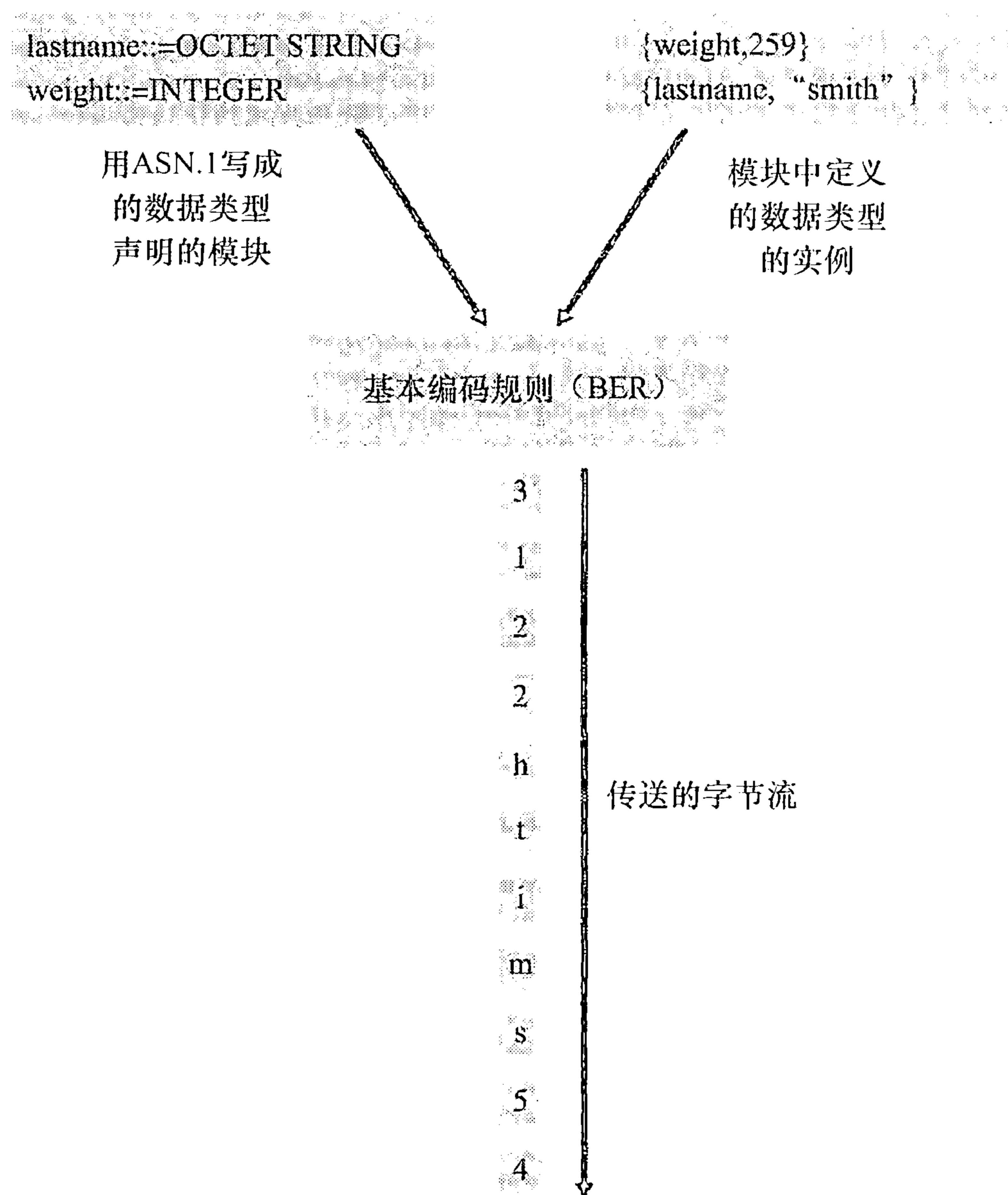


图 9-9 BER 编码例子

在以上的讨论中，我们仅仅接触了 ASN.1 的少而简单的内容。如要学习更多 ASN.1，可参考的资料包括 ASN.1 标准文档 [ISO X.680 2002]、在线 OSI 相关书籍 [Larmouth 2012] 和 ASN.1 相关的 Web 站点 [OSS 2012] 和 [OID Repository 2012]。

9.5 小结

我们对网络管理以及所有与网络有关的内容的学习现在已经结束！

在有关网络管理的最后一章中，我们以需求为动机，为网络管理员提供适当的工具，以监视、测试、轮询、配置、分析、评估和控制网络的运营。网络管理员的责任是保持网络“启动并运行良好”。我们与复杂系统如发电厂、飞机和人类组织的管理进行类比，有助于启发这种需求。我们看到网络管理系统的体系结构围绕 5 个关键组件：①一个网络管

理者；②一组被管的远程（距网络管理者）设备；③在这些设备中包含有关这些设备的状态和操作数据的管理信息库（MIB）；④报告 MIB 信息并在网络管理者控制下采取措施的远程代理；⑤网络管理者和远程设备之间的通信协议。

接着我们深入研究了因特网标准管理框架的细节，特别是 SNMP 协议。我们看到了 SNMP 是如何对网络管理体系结构的 5 个关键组件实例化的，花了不少时间来研究 MIB 对象、SMI（用于定义 MIB 的数据定义语言）和 SNMP 协议本身。注意到 SMI 和 ASN.1 是密不可分的，ASN.1 在 ISO/OSI 7 层参考模型的表示层中起着关键作用，我们则简要地学习了 ASN.1。也许比 ASN.1 自身细节更为重要的是，在网络中机器特定数据格式之间的转换。尽管某些网络体系结构由于具有表示层而明确声明了这种服务的重要性，但在因特网协议栈中缺少这个层次。

还值得一提的是，本书还有许多重要的网络管理主题我们没有涉及，例如，故障识别和管理，前瞻性异常检测，告警关联和服务管理的更大的问题（例如，与网络管理形成对照）。这些主题虽然重要，但对它们的讨论应当由专门的文献来进行，我们提请读者参考 9.1 节中提到的文献。

课后习题和问题



复习题

9.1 节

- R1. 为什么网络管理员得益于使用网络管理工具？描述 5 种相关情况。
- R2. ISO 定义的网络管理的 5 个领域是什么？
- R3. 网络管理和服务管理的主要区别是什么？

9.2 节

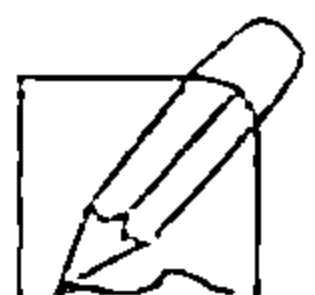
- R4. 定义下列术语：管理实体，被管设备，管理代理，MIB，网络管理协议。

9.3 节

- R5. 网络管理中 SMI 的作用是什么？
- R6. SNMP 中的请求响应报文和陷阱报文之间的重要区别是什么？
- R7. SNMP 中使用的 7 种报文类型是什么？
- R8. “SNMP 引擎”的含义是什么？

9.4 节

- R9. ASN.1 对象标识树的目的的是什么？
- R10. 在 ISO/OSI 参考模型的表示层中，ASN.1 的作用是什么？
- R11. 因特网具有表示层吗？如果没有，如何关注所涉及的机器体系结构的差异（例如，在不同机器上整数的不同标识）？
- R12. TLV 编码意味着什么？



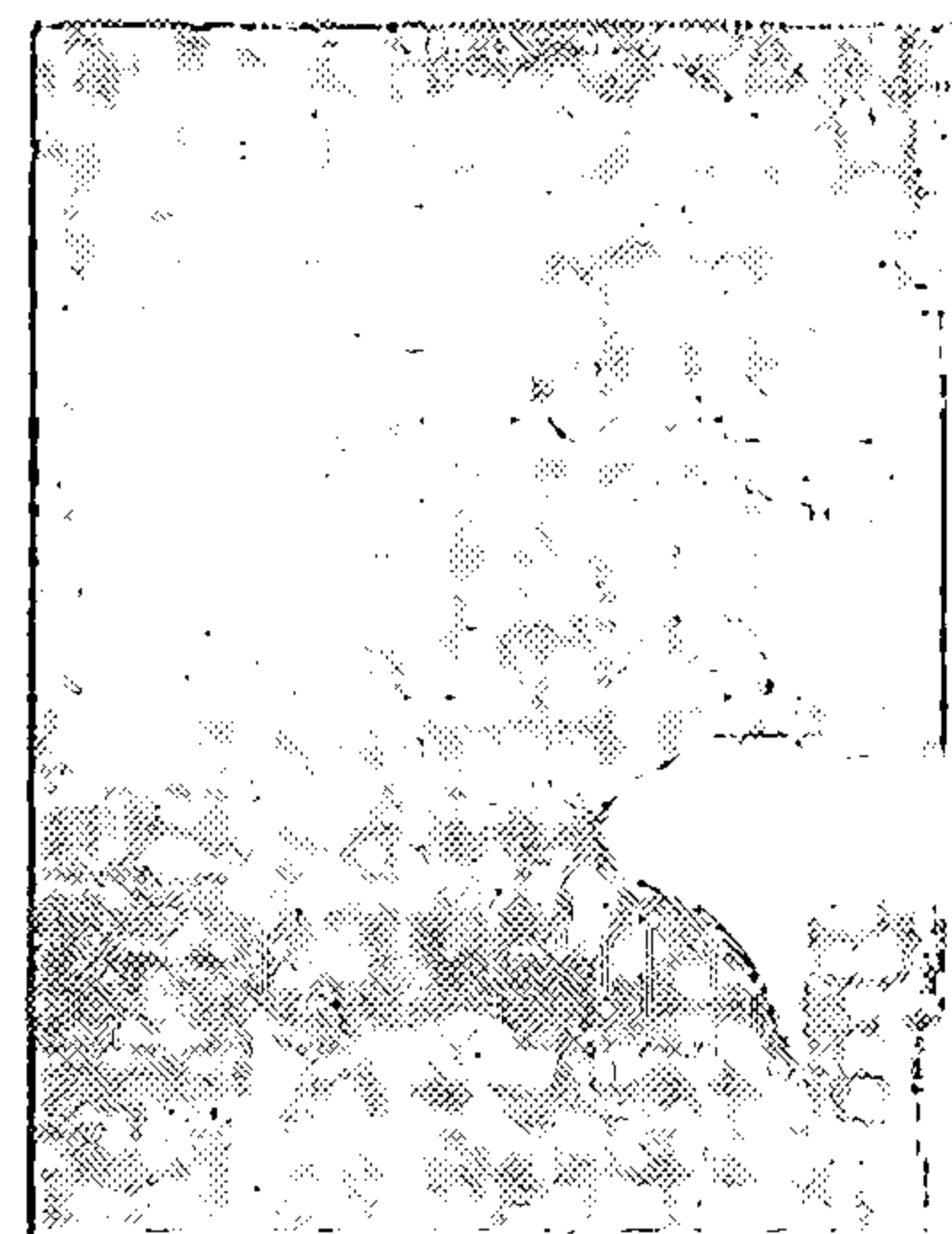
习题

- P1. 考虑在管理实体和被管设备之间发生通信的两种方式：请求响应方式和陷阱方式。从以下方面考虑这两种方式的优缺点：（1）开销；（2）当异常事件出现时通知的时间；（3）有关管理实体和设备之间丢失报文的健壮性。

- P2. 在 9.3 节中我们看到，用不可靠的 UDP 数据报传输 SNMP 报文是更可取的方式。请你阐述 SNMP 设计者选择 UDP 而不是 TCP 作为 SNMP 运输协议的理由。
- P3. ICMP 协议的 ASN.1 对象标识符是什么（参见图 9-3）？
- P4. 假设你在为一家美国公司工作，该公司希望开发自己的 MIB 来管理生产线。它将在对象标识树（图 9-3）的什么地方注册？（提示：为了回答这个问题，你需要钻研 RFC 或其他文档。）
- P5. 9.3.2 节讲过，一家私营公司（企业）能够在专用分支 1.3.6.1.4 下面创建自己的 MIB 变量。假定 IBM 公司要为它的 Web 服务器软件创建一个 MIB。在 1.3.6.1.4 后面的下一个 OID 限定词是什么？（为了回答这个问题，你需要参考 [IANA 2009b]。）搜索 Web，看看是否能找到存在这样用于 IBM 服务器的 MIB。
- P6. 在 TLV 编码中，为什么你认为长度位于值之前（而不是长度跟在值之后）？
- P7. 考虑图 9-9。|weight, 165| |lastname, "Michael"| 的 BER 编码是什么？
- P8. 考虑图 9-9。|weight, 145| |lastname, "Sridhar"| 的 BER 编码是什么？

人物专访

Jennifer Rexford 是普林斯顿大学计算机科学系的教授。她的研究具有宽广的目标：通过特别强调路由选择协议，使得计算机网络更容易设计和管理。从 1996 年到 2004 年，她是 AT&T 实验室研究网络管理和性能部的成员。在 AT&T 期间，她设计用于网络测量、流量工程和 AT&T 主干网络中部署的路由器配置的技术和工具。Jennifer 是《Web 协议和实践：网络协议、缓存和流量测量》（Web Protocols and Practice: Networking Protocols, Caching, and Traffic Measurement）一书的合作者，该书于 2001 年 5 月由 Addison-Wesley 出版社出版。她在 2003 ~ 2007 年间担任 ACM SIGCOMM 的主席。她于 1991 年从普林斯顿大学获得电子工程工程理学士（BSE）学位，分别于 1993 年和 1996 年从密歇根大学获得电子工程工程硕士（MSE）和计算机科学博士（PhD）学位。2004 年，Jennifer 成为 ACM 的 Grace Murray Hopper 奖杰出青年计算机专家，她的名字出现在麻省理工学院“技术展望 100 名”35 岁以下顶级创新家名单上。



Jennifer Rexford

- 请描述在您职业生涯中所从事的最令人兴奋的一两个项目。最大的挑战是什么？

当我在 AT&T 做研究人员的时候，我们一群人设计一种新方法，用于管理因特网服务提供商主干网中的路由选择。传统上，网络管理员逐台配置路由器，这些路由器运行分布式协议以通过网络计算出路径。我们认为如果网络管理员能够运用基于拓扑和流量的网络范围视图直接控制路由器转发流量的方式，网络管理将变得更为简单和灵活。我们设计并建造的路由选择控制平台（RCP）能够在单台商用计算机上为所有 AT&T 主干计算路由，并且能够不加修改地控制原有的路由器。对我而言，这个项目是令人兴奋的，因为我们有了一个另类的想法、一个工作的系统并且最终在一个运营网络中进行了实际部署。

- 未来在网络管理领域中您将预见到什么变化和创新？

不仅仅是在现有网络上拧紧网络管理的螺栓，研究人员及从业人员正在开始设计本质上更容易管理的网络。就像我们早期在 RCP 上的工作，在所谓软件定义网络（Software Defined Networking, SDN）中的主要思想是运行控制器，控制器能够在使用标准协议的基础交换机中安装低层次处理分组的规则。这种控制器能够运行各种网络管理应用程序，诸如动态访问控制、无缝用户移动性、流量工程、服务器负载均衡和能量高效网络等等。我相信通过重新思考网络设备和管理它们的软件之间的关系，SDN 增大了彻底搞清网络管理的机会。

- 您认为网络的未来以及因特网的路在何方？

网络是一个令人兴奋的领域，因为应用和支撑技术一直都在改变。我们总是在改造自己！有谁在甚

至 5 年或 10 年前能预测到智能手机主宰的局面，允许移动用户接入现有应用以及基于位置的新服务呢？云计算的出现从根本上改变了用户和他们所运行的应用之间的关系，联网的传感器能够使大量的新型应用成为可能。创新的步伐真正鼓舞人心。

基础网络是所有这些创新的要素。的确，网络是众所周知的“绊脚石”，即限制性能、降低可靠性、限制应用以及使服务的部署和管理复杂化。我们应当使未来的网络朝着如同我们呼吸的空气一样不可见的方向进行努力，这样它就决不会阻碍新的思想和有价值的服务了。为了做到这一点，我们需要在网络设备和协议（以及随之而来的首字母缩写词）之上提升抽象水平，使得我们能够将网络作为一个整体来思考。

◦ 什么人启发您的职业灵感？

长期以来，我在国际计算机科学学会（International Computer Science Institute）受到 Sally Floyd 的启迪。她的研究总是有明确目标，关注因特网面临的重大挑战。她深入钻研难题直到完全理解该问题和解空间，并且她将大量精力投入到“眼见为实”中，例如将她的思想注入协议标准和网络设备中。此外，她通过在大量的标准和研究组织中的专业服务，以及通过创建工具（如广泛使用的 ns-2 和 ns-3 模拟器）使其他研究人员能够取得成功，而回馈社区。她于 2009 年退休，但她对该领域的影响将持续存在。

◦ 对于以计算机科学和网络为职业的学生，您的忠告是什么？

网络本质上是一个跨学科领域。将其他学科技术应用到网络问题是推动本领域向前发展的很好方式。我们已经看到来自不同领域的知识为网络带来的巨大突破，这些领域诸如排队论、博弈论、控制论、分布式系统、网络优化、编程语言、机器学习、算法、数据结构等。我认为逐步精通相关领域，或与那些领域的专家密切合作是一种为网络构建更强大基础的极好方式，这样我们能够学会如何构建值得社会信任的网络。网络是令人兴奋的领域，除了它是理论学科外，还因为我们创造现实中人们正在使用的实际人工制品。掌握如何设计和建造系统，即通过增长操作系统、计算机体系结构等方面的经验，是另一种增强你的网络知识、进而改变世界的了不起方式。

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有关 URL 的注释。在下面的引用中，我们提供了 Web 网页、仅在 Web 上有的文档和没有被会议或杂志出版的其他材料的 URL（当我们能够指出这些材料的位置时）。我们没有提供有关会议和杂志出版物的 URL，因为这些文档通常能够通过如下方式找到：使用某个搜索引擎，经该会议的 Web 站点（例如在所有 ACM SIGCOMM 会议和专题讨论会中的文章能够通过 <http://www.acm.org/sigcomm> 找到），或通过订阅数字图书馆。尽管到 2012 年 1 月，下面提供的所有 URL 都是有效的，但 URL 可能会过期。对于过期的文献，请参考本书的在线版本（<http://www.awl.com/kurose-ross>）。

有关因特网请求评论（RFC）的注释。因特网 RFC 的副本在多个网站上都可供使用。因特网协会（监管 RFC 文档的组织）的 RFC 编辑们维护着网站 <http://www.rfc-editor.org>。该网站允许你通过标题、编号或作者来搜索某个特定的 RFC 文档，并将显示出对任何所列 RFC 的更新。因特网 RFC 可以被后面的 RFC 所更新或淘汰。我们喜欢的获取 RFC 文档的网站是初始 RFC 源，即 <http://rfc-editor.org>。

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